

Analyzing an Experiment: Difference of Proportions

MATH 213

Group 1

The table below shows the actual experimental data from the promotion-decision experiment that we studied earlier in the semester.

	“Male” file	“Female” file
Promote	21	14
Did not promote	3	10

What parameter describing the 48 bank administrators do we want to estimate?

~~Difference in promotion rate based on gender~~

The difference in proportions of those who would promote if all study participants were given a male file and if all study participants were given a female file

What point estimate for the parameter does the study data give us?

$$21/24 - 14/24 = 0.875 - 0.583 = 0.292$$

What is the estimated standard error? Show some of your work.

$$SE_m = \sqrt{(0.875(1-0.875)/_{24})} = 0.068$$

$$SE_f = \sqrt{(0.583(1-0.583)/_{24})} = 0.101$$

$$SE_{diff} = \sqrt{(SE_m^2 + SE_f^2)} = 0.121$$

Use your estimated standard error to create a 95% confidence interval for the parameter. Write a sentence interpreting your 95% interval.

$$0.292 \pm 0.121 * 1.96 = (0.055, 0.529)$$

We are 95% confident that the difference in proportion between males and females receiving a promotion is between 0.055 and 0.529, in favor of males. That is, if every supervisor was given a male and female file, between 5.5% and 52.9% of males would be promoted more than females.

We are 95% confident that the proportion of promotions among managers given male files is between 0.055 and 0.529 higher than managers given female files.

Group 2

The table below shows the actual experimental data from the promotion-decision experiment that we studied earlier in the semester.

	“Male” file	“Female” file
Promote	21	14
Did not promote	3	10

What parameter describing the 48 bank administrators do we want to estimate? If bank administrators as a whole are more likely to promote males over females. Or basically, the promotion based on gender.

What point estimate for the parameter does the study data give us? $21/24 = .875$ $14/24 = .583$
 $.875 - .583 = .292$

What is the estimated standard error? Show some of your work.

$$\sqrt{((.875(1-.875))/24)+((.583(1-.583))/24))}=0.121189589762$$

Confidence interval

$$0.242379179524+.292=0.534379179524$$

$$.292- 0.242379179524=0.049620820476$$

Use your estimated standard error to create a 95% confidence interval for the parameter. Write a sentence interpreting your 95% interval.

We are 95% confident that the interval of bank administrators who favor promoting males over those favoring females resides between the percentages of 4.96% and 53.44%. That is, the range of gender bias in promotion is averagely between 4.96% and 53.44%

Group 3

The table below shows the actual experimental data from the promotion-decision experiment that we studied earlier in the semester.

	“Male” file	“Female” file
Promote	21	14
Did not promote	3	10

What parameter describing the 48 bank administrators do we want to estimate?

The difference in how often men get promoted vs. how often women get promoted.

What point estimate for the parameter does the study data give us?

$21/24 = 0.875$

$14/24 = 0.583$

$0.875 - 0.583 = 0.292$ difference in promotion between men and women.

What is the estimated standard error? Show some of your work.

$$\mathbf{SE_M = 0.068}$$

$$\mathbf{SE_F = 0.101}$$

Use your estimated standard error to create a 95% confidence interval for the parameter. Write a sentence interpreting your 95% interval.

Group 4

The table below shows the actual experimental data from the promotion-decision experiment that we studied earlier in the semester.

	“Male” file	“Female” file
Promote	21	14
Did not promote	3	10

What parameter describing the 48 bank administrators do we want to estimate?

The difference in proportion of those who would promote if all participants were given a male file and all participants were given a female file.

What point estimate for the parameter does the study data give us?

$$21/24 - 14/24 = 7/24 = .29$$

What is the estimated standard error? Show some of your work.

$$SE_m = 0.0675$$

$$SE_f = 0.101$$

$$Geomean = ((SE_m)^2 + (SE_f)^2)^{.5} = 0.121$$

Use your estimated standard error to create a 95% confidence interval for the parameter. Write a sentence interpreting your 95% interval.

$$.29 \pm 2 \cdot .121 = (0.048, 0.532)$$

We are 95% confident that the difference in proportion of those who would promote if all participants were given a male file and all participants were given a female file will be found between 4.8% and 53.2%.

Group 5

The table below shows the actual experimental data from the promotion-decision experiment that we studied earlier in the semester.

	“Male” file	“Female” file
Promote	21	14
Did not promote	3	10

What parameter describing the 48 bank administrators do we want to estimate?

The difference in proportions of bank administrators that received promotions among male and females.

What point estimate for the parameter does the study data give us?

$$21/24 = 0.875 \quad 14/24 = 0.583 \quad \longrightarrow \quad 0.875 - 0.583 = 0.292$$

0.292

What is the estimated standard error? Show some of your work.

$$SE_{\text{male}} = \sqrt{0.875(1-0.875)/24} = 0.0675$$

$$SE_{\text{female}} = \sqrt{0.583(1-0.583)/24} = .101$$

Geomean

$$SE_{\text{diff}} = \sqrt{SE^2 + SE^2}$$

$$\rightarrow \sqrt{0.0675^2 + .101^2} = .121$$

Use your estimated standard error to create a 95% confidence interval for the parameter. Write a sentence interpreting your 95% interval.

Sample proportion $\pm Z^*(SE)$

$$.292 + (2 * .121) = 0.534$$

$$.292 - (2 * .121) = 0.05$$

We are 95% confident that the populations proportion difference in females and males receiving promotions is between 0.05 and 0.534

Group 6

The table below shows the actual experimental data from the promotion-decision experiment that we studied earlier in the semester.

	“Male” file	“Female” file
Promote	21	14
Did not promote	3	10

What parameter describing the 48 bank administrators do we want to estimate?

Difference in proportions of bank administrators that were promoted between the male and female files.

What point estimate for the parameter does the study data give us?

Proportion of male promotions: 0.875

Proportion of female promotions: 0.583

Difference: 0.292

What is the estimated standard error? Show some of your work.

Proportion of male promotions: 0.875

Proportion of female promotions: 0.583

SE for male files: $\text{sqrt}((0.875 * (1-0.875)) / 24) = \underline{0.068}$

SE for female files: $\text{sqrt}((0.583 * (1-0.583)) / 24) = \underline{0.101}$

SE for difference in proportion: $\text{sqrt}(0.068^2 + 0.101^2) = \underline{0.122}$

Use your estimated standard error to create a 95% confidence interval for the parameter. Write a sentence interpreting your 95% interval.

Formula: [statistic] $\pm$ [z*] * [SE]

Lower bound: $0.292 - (2 * 0.122) = \underline{0.048}$

Upper bound: $0.292 + (2 * 0.122) = \underline{0.536}$

95% CI: (0.048, 0.536)

We are 95% confident that, in the population, the proportion of promotions among those given male files is between 0.05 - 0.54 higher than those who were given female files.

Group 7

The table below shows the actual experimental data from the promotion-decision experiment that we studied earlier in the semester.

	“Male” file	“Female” file
Promote	21	14
Did not promote	3	10

What parameter describing the 48 bank administrators do we want to estimate?
Difference in proportions of male vs. females who were promoted

What point estimate for the parameter does the study data give us?
Difference in proportions (0.292)

What is the estimated standard error? Show some of your work.

$$\sqrt{p(1-p)/n}$$

$$\sqrt{0.292(1-0.292)/48}$$

$$0.0656$$

Use your estimated standard error to create a 95% confidence interval for the parameter. Write a sentence interpreting your 95% interval.

$$(0.1608, 0.4232)$$

We are 95% confident that the true difference in males vs. females promoted is between 16% - 42%.

Group 8

The table below shows the actual experimental data from the promotion-decision experiment that we studied earlier in the semester.

	“Male” file	“Female” file
Promote	21	14
Did not promote	3	10

What parameter describing the 48 bank administrators do we want to estimate?

What point estimate for the parameter does the study data give us?

What is the estimated standard error? Show some of your work.

Use your estimated standard error to create a 95% confidence interval for the parameter. Write a sentence interpreting your 95% interval.

Group 9

The table below shows the actual experimental data from the promotion-decision experiment that we studied earlier in the semester.

	“Male” file	“Female” file
Promote	21	14
Did not promote	3	10

What parameter describing the 48 bank administrators do we want to estimate?

The difference in proportions of males who got promoted vs females

What point estimate for the parameter does the study data give us?

If the male or female actually got promoted

$$21/24 - 14/24 = 0.292$$

What is the estimated standard error? Show some of your work.

$$SE_{\text{male}} = \sqrt{(0.875(1 - 0.875)) / 24} = 0.0675$$

$$SE_{\text{female}} = \sqrt{(0.5833(1-0.5833)) / 24} = 0.1006$$

$$SE_{\text{diff}} = \sqrt{0.0675^2 + 0.1006^2} = 0.1211$$

Use your estimated standard error to create a 95% confidence interval for the parameter. Write a sentence interpreting your 95% interval.

$$95_CI = 0.292 \pm 2 * 0.1211 = (0.0498, 0.5342)$$

We are 95% confident that the true difference in proportion of people who would promote if all study participants were given a male file and if all study participants were given a female file is between 4.95% and 53.42%

Group 10

The table below shows the actual experimental data from the promotion-decision experiment that we studied earlier in the semester.

	“Male” file	“Female” file
Promote	21	14
Did not promote	3	10

What parameter describing the 48 bank administrators do we want to estimate?

The difference in proportions of those who would promote if all study participants were given a male file and if all study participants were given a female file

What point estimate for the parameter does the study data give us?
Whether the female or male employee was getting promoted or not

What is the estimated standard error? Show some of your work.

$$SE_m = \sqrt{(.437(1-.437))/48} = 0.0477$$

$$\sqrt{(.291(1-.291))/48} = 0.0711$$

$$\sqrt{se_1^2 + se_2^2} = 0.0856$$

Use your estimated standard error to create a 95% confidence interval for the parameter. Write a sentence interpreting your 95% interval.

$$1.96 * .0856 = .16776$$

$$.541 + .16776 = .70876$$

$$.541 - .16776 = .37324$$

We are 95% confident that the proportion of males and females that didn't get hired is between 37.32% and 70.87%

Group 11

The table below shows the actual experimental data from the promotion-decision experiment that we studied earlier in the semester.

	“Male” file	“Female” file
Promote	21	14
Did not promote	3	10

What parameter describing the 48 bank administrators do we want to estimate?

Difference in the rate of promotion between the genders

What point estimate for the parameter does the study data give us?

$$21/24 - 14/24 = 0.875 - 0.583 = 0.292$$

What is the estimated standard error? Show some of your work.

$$0.294/\sqrt{24}=se$$

$$se=0.06001249869$$

Use your estimated standard error to create a 95% confidence interval for the parameter. Write a sentence interpreting your 95% interval.

$$1.96*0.06001249869=0.11762449743$$

$$35/48=0.729166666666$$

$$0.729166666666+-0.11762449743=0.84679116409, 0.61154216923$$

This means that we are 95% confident that the mean difference should fall between 61%-84%